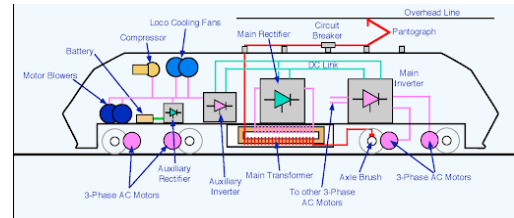
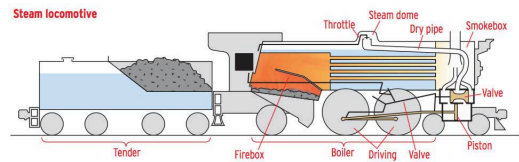


Homework problem draft:

The government of California is working on building a High Speed Rail (HSR) network that connects Los Angeles and San Francisco, as well as other major cities along the way. The government has hired you (yes, you!) to see if this is a better solution. Your goal is to compare HSR to a petroleum-powered train.

For most trains around the world, the fuel conversion happens in a train car called the “locomotive”, where diesel or petroleum is burned to make the train move forward. HSR is controlled by an Electric Multiple Unit (EMU), where each carriage helps propel the train and is powered by electricity (usually through a third rail at the bottom or an overhead wire).

Let’s look at a single locomotive (Top) and a EMU carriage (Bottom). We want to know, for each carriage,



- a) Is it a closed or open system?
 - i) Locomotive: Open system, mass escapes through system boundary via emissions.
 - ii) EMU: Closed system (?), no mass crosses the system boundary and power is contained internally within the system and from the external power storage (third rail/overhead line).
- b) What type of energy conversions happen in the system?
 - i) Locomotive:
 - 1) Chemical $\rightarrow$ Mechanical
 - 2) Chemical $\rightarrow$ Thermal
 - ii) EMU:
 - 1) Electrical $\rightarrow$ Mechanical

Now let’s look at each of the systems individually.

A boiler produces saturated water vapor (steam) at 1.4 MPa. This steam flows through a dry pipe to the steam dome, where it is admitted into the locomotive's piston-cylinder assembly.

Process Description:

- State 1 (Piston Inlet): Saturated vapor enters the piston at a constant pressure of 1.4 MPa.
- State 2 (Piston Exhaust): After expanding against the piston, the steam is exhausted as a saturated mixture at 400 kPa.
- The mass flow rate of steam through the piston is 1×10^{-4} kg/s.

- The piston work conversion process operates at 50% efficiency.

c) Calculate the actual power output (in Watts) delivered by the piston.

④ c) We know $P_1 = 1.4 \text{ MPa}$ and $x_1 = 1$. We also know $P_2 = 400 \text{ kPa}$ and $\dot{m} = 1 \cdot 10^{-4} \text{ kg/s}$. We can use table A-5 to find T_1 and T_2 .
 $T_1 = 195.04^\circ\text{C}$ and $T_2 = 143.61^\circ\text{C}$.
 We know by our first law equation,
 $\frac{dE}{dt} = \dot{Q} + \dot{W} + \Delta \dot{m} \theta$,
 assuming steady state, $\frac{dE}{dt} = 0$.
 $0 = \dot{W}_{in} - \dot{W}_{out} + \dot{m} \left(\cancel{\frac{1}{2} v^2} + \cancel{pe} + h_1 - h_2 \right)$
 $0 = -\dot{W}_{out} + \dot{m} (h_1 - h_2)$
 $-\dot{m} (h_1 - h_2) \quad -\dot{m} (h_1 - h_2)$
 $\dot{m} (h_2 - h_1) = -\dot{W}_{out}$
 $\dot{m} C_p \Delta T = -\dot{W}_{out}$ assuming an ideal gas.
 We know $\dot{m} = 1 \cdot 10^{-4} \text{ kg/s}$, $C_p = 1.8723 \text{ kJ/kgK}$,
 and $\Delta T = 143.61^\circ\text{C} - 195.04^\circ\text{C} = -51.43 \text{ K}$.
 Thus, $-\dot{W}_{out} = -51.43 \text{ K} (1.8723 \text{ kJ/kgK}) (1 \cdot 10^{-4} \text{ kg/s})$
 $-\dot{W}_{out} = -0.00963 \text{ kJ/s} = -9.63 \text{ J/s}$
 $\dot{W}_{out, id} = 9.63 \text{ J/s} = 9.63 \text{ W}$
 continue.

④ c)

$$\frac{\dot{W}_{act}}{\dot{W}_{id}} = 0.5$$

$$\left(\frac{\dot{W}_{act}}{9.63W} = 0.5 \right) 9.63W$$

$$\boxed{\dot{W}_{act} = 4.81W}$$

For the EMU, the input energy source is electricity powered by rails.

- d) If electric power enters the rail at 2.43 MW and operates at 75% efficiency, how much power is produced by a single carriage?

d) We know $\eta = \frac{\dot{W}_{act}}{\dot{W}_{id}}$

We know $\dot{W}_{id} = 2.43 \text{ MW}$ and $\eta = 0.75$.

Thus, $\left(0.75 = \frac{\dot{W}_{act}}{2.43 \text{ MW}} \right) 2.43 \text{ MW}$

$$\boxed{1.82 \text{ MW} = \dot{W}_{act}}$$

Finally, let's see how fast each of the trains can go. For regular trains, the locomotive "pulls" the other train cars. Assume the locomotive weighs 200,000 kgs and each of the 40 train cars weighs 130,000 kgs (including people), so the one locomotive pulls 40 cars.

- e) Based on the results of part c), determine the speed of the full diesel train.

Each carriage in an EMU produces the same amount of work, so they don't need to be "pulled" by a main carriage.

$$\dot{W}_{act} = 4.81W$$

$$P_{tot} = \dot{W}_{act} + 1820000W = 1820004.81W$$

$$g = 9.81 \text{ m/s}^2$$

$$M = 200,000 \text{ kg}$$

$$c_r = 0.002 \text{ (typical coefficient of rolling resistance for steel train)}$$

$$F = c_r * M * g = 0.002 * 200000 \text{ kg} * 9.81 \text{ m/s}^2 = 3924 \text{ N}$$

$$P_{tot} = F * v$$

$$v = \frac{P_{tot}}{F}$$

$$v = \frac{1820004.81W}{3924 N}$$

$$v = 463.81 m/s$$

- f) If there are 20 carriages that each weigh 130,000 kgs, how fast does the EMU go?

$$\dot{W}_{act} = 4.81W$$

$$P_{tot} = \dot{W}_{act} + 1820000W = 1820004.81W$$

$$g = 9.81 m/s^2$$

$$M = 200,000kg + 20(130,000kg) = 2800000 kg$$

$$c_r = 0.002 \text{ (typical coefficient of rolling resistance for steel train)}$$

$$F = c_r * M * g = 0.002 * 2800000 kg * 9.81 m/s^2 = 54,936 N$$

$$P_{tot} = F * v$$

$$v = \frac{P_{tot}}{F}$$

$$v = \frac{1820004.81W}{54,936 N}$$

$$v = 33.13 m/s$$